

Data Loading (ETL)

Assignment



Order_ID	Customer_ID	Sales_Amount	Order_Date
O101	C001	4500	12-01-2024
O102	C002	Null	15-01-2024
O103	C003	3200	2024/01/18
O101	C001	4500	12-01-2024
O104	C004	Three Thousand	20-01-2024
O105	C005	5100	25-01-2024
O106	C006	6200	26-01-2024
O107	C007	7200	27-01-2024
O108	C008	NULL	2024/01/28
O109	C009	3900	29-01-2024
O110	C010	4500	30-01-2024
O111	C011	5400	31-01-24
O112	C012	4800	01-02-2024
O113	C013	5200	2024/02/02
O114	C014	Five Thousand	03-02-2024
O115	C015	6100	04-02-2024
O116	C016	7000	05-02-24
O117	C017	NULL	06-02-2024
O118	C018	3400	2024/02/07
O119	C019	8900	08-02-2024
O120	C020	7600	09-02-2024
O121	C021	5100	10-02-2024
O122	C022	4300	11-02-24
O123	C023	2800	2024/02/12
O124	C024	9900	13-02-2024
O125	C025	4100	14-02-2024
O126	C026	NULL	15-02-2024

Order_ID	Customer_ID	Sales_Amount	Order_Date
O127	C027	3600	2024/02/16
O128	C028	5400	17-02-2024
O129	C029	Seven Thousand	18-02-2024
O130	C030	6200	19-02-24
O131	C031	5800	20-02-2024
O132	C032	4700	21-02-2024
O133	C033	3100	2024/02/22
O134	C034	NULL	23-02-2024
O135	C035	4500	24-02-2024
O136	C036	5300	25-02-2024
O120	C020	7600	09-02-2024
O138	C038	6800	26-02-2024
O138	C038	7200	27-02-24

Use this dataset for following questions

Q1. Data Understanding

Identify all data quality issues present in the dataset that can cause problems during data loading.

Q2. Primary Key Validation

Assume Order_ID is the Primary Key.

- Is the dataset violating the Primary Key rule?
- Which record(s) cause this violation?

Q3. Missing Value Analysis

Which column(s) contain missing values?

- List the affected records
- Explain why loading these records without handling missing values is risky

Q4. Data Type Validation

Identify records where Sales_Amount violates expected data type rules.

- Which record(s) will fail numeric validation?
- What would happen if this dataset is loaded into a SQL table with Sales_Amount as DECIMAL?

Q5. Date Format Consistency

The **Order_Date** column has multiple formats.

- a) List all date formats present in the dataset
- b) Why is this a problem during data loading?

Q6. Load Readiness Decision

Based on the dataset condition:

- a) Should this dataset be loaded directly into the database? (Yes/No)
- b) Justify your answer with at least three reasons

Q7. Pre-Load Validation Checklist

List the exact pre-load validation checks you would perform on this dataset before loading.

Q8. Cleaning Strategy

Describe the step-by-step cleaning actions required to make this dataset load-ready.

Q9. Loading Strategy Selection

Assume this dataset represents daily sales data.

- a) Should a Full Load or Incremental Load be used?
- b) Justify your choice.

Q10. BI Impact Scenario

Assume this dataset was loaded without cleaning and connected to a BI dashboard.

- a) What incorrect results might appear in Total Sales KPI?
- b) Which records specifically would cause misleading insights?
- c) Why would BI tools not detect these issues automatically?